Youqi Liao



First-Year PhD Student | Vision/Language Localization · Place Recognition · Sensor Fusion

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Education

Wuhan University — Wuhan, China

M.Sc. in Photogrammetry and Remote Sensing — Sep 2023 – Jun 2026

- Publications: Published five first-author papers in top-tier SCI Q1 journals.
- Honors and Awards: National Scholarship, BYD Scholarship, and First-Class Academic Scholarship.

Wuhan University — Wuhan, China

B.Eng. in Surveying Engineering — Sep 2019 – Jun 2023

- Thesis: An Online LiDAR–Camera Calibration Algorithm Based on Semantic Edge Features.
- GPA: 3.71/4.00.
- Honors and Awards: Chen Yongling Scholarship and Wuhan University Second-Class Scholarship.

Research Experience

Graduate Research Projects — Wuhan, China

State Key Laboratory (LIESMARS), Wuhan University — Sep 2023 – Jun 2026

- Mobile-Seed: Joint semantic segmentation and boundary detection for mobile robots.
- CoFiI2P: Coarse-to-fine image-to-point-cloud registration via cross-modal correspondences.
- OPAL: Visibility-aware point-cloud-to-navigation-map place recognition.
- OSMLoc: Semantics- and geometry-guided single-image localization in OpenStreetMap.
- TOL: Text-to-OpenStreetMap localization from natural-language descriptions.

Undergraduate Research and Innovation Project — Wuhan, China

State Key Laboratory (LIESMARS), Wuhan University — Sep 2021 – Jun 2023

- SE-Calib: Semantic edge-based online LiDAR–camera calibration in urban road scenes.

Publications

*: Equal contribution

[1] Y. Liao, J. Li, S. Kang, et al. SE-Calib: Semantic Edge-Based LiDAR–Camera Boresight Online Calibration in Urban Scenes. IEEE TGRS. (Top SCI Q1 Journal)

[2] Y. Liao, S. Kang, J. Li, et al. Mobile-Seed: Joint Semantic Segmentation and Boundary Detection for Mobile Robots. IEEE RA-L. (Top SCI Q2 Journal)

[3] S. Kang*, Y. Liao*, J. Li, et al. CoFiI2P: Coarse-to-Fine Correspondences-Based Image-to-Point-Cloud Registration. IEEE RA-L. (Top SCI Q2 Journal)

[4] S. Kang*, Y. Liao*, Y. Xia, et al. OPAL: Visibility-Aware LiDAR-to-OpenStreetMap Place Recognition via Adaptive Radial Fusion. Conference on Robot Learning, 2025. (Top-Tier Robotics Conference)

[5] Y. Liao*, X. Chen*, S. Kang, et al. OSMLoc: Single Image-Based Visual Localization in OpenStreetMap with Fused Geometric and Semantic Guidance. Information Fusion. (Top SCI Q1 Journal)

[6] S. Kang, Y. Liao, P. Wang, et al. VLM-Loc: Localization in Point Cloud Maps via Vision-Language Models. CVPR 2026. (CCF Class A Conference)

[7] Y. Liao, S. Kang, J. Xu, et al. TOL: Textual Localization with OpenStreetMap. IEEE RA-L. (Top SCI Q2 Journal)